The change of variable from Schrodinger to QFT Document

The value of R is a constant distance

The onduscular equation $\frac{2m}{\hbar^2}(E-U(r))rf(r,t)+\frac{R}{r}\frac{\partial^2(rf(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 rf(r,t)}{\partial t^2}$ (1) is valid in both spherical and cylindrical coordinates; it emerges from a combination of Schrodinger, which is the first degree with respect to time, and the Klein-Gordon temporal equation of the second degree with respect to time. Also, we chose a parameter R, thus for a small radius $r_{Bohr,}$ for the atomic value to coincide with the Schrodinger equation exactly and predict the evolution of the wave function to huge values of r about the galactic distances. The wave function of the Schrodinger equation has the supposition that is null at infinity, is normed, has the dimension of $[L^{-3/2}]$, and predicts the probability of the existence of the microparticle in volume dV. Schrodinger's equation is (2):

$\frac{2m}{\hbar^2}(E-U(r))f(r,t)+\frac{\partial^2(f(r,t))}{\partial r^2}=ih\frac{\partial f(r,t)}{\partial t}$ & Klein-Gordon is $\frac{2m}{\hbar^2}f(r,t)+\frac{\partial^2(f(r,t))}{\partial r^2}=\frac{1}{c^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ (3)

Thus, Eq. (1) for m = 0 is a wave equation with variable parameters and seems to be true. $\frac{R}{r}\frac{\partial^2(rf(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 rf(r,t)}{\partial t^2}$ (4) But all 3 equations are different, and the wave equation is not satisfactory

$$\frac{\partial^2(f(r,t))}{\partial r^2}=\frac{1}{c^2}\frac{\partial^2 f(r,t)}{\partial t^2} \quad (5),$$

1.1 Step 1 — Energy balance for a bound electron

For a hydrogenic electron in a Coulomb field, the non-relativistic energy is $E=\frac{1}{2}m_eV^2-\frac{e^2}{4\pi\varepsilon_0 r}+\frac{l(l+1)\hbar^2}{2m_e r^2}$. (1.1)

Solving for the kinetic term and dividing by $\hbar^2/(2m_e)$ gives the "wavenumber-squared" combination

$$\frac{2m_e}{\hbar^2}\left[E+\frac{e^2}{4\pi\varepsilon_0 r}-\frac{l(l+1)\hbar^2}{2m_e r^2}\right]=\frac{m_e^2V^2}{\hbar^2}=k^2(r). \quad (1.2)$$

1.2 Step 2 — Onduscular wave ansatz

In TauTimp, an electron is represented by a spatial-temporal wave $f(r,t)$ propagating at the local "ondoscopic" speed V, with the cosmological scale R entering through a curvature-corrected Laplacian. The local d'Alembertian satisfies

$$\Box f=\frac{R}{r^2}\frac{\partial^2(rf)}{\partial r^2}-\frac{1}{V^2}\frac{\partial^2 f}{\partial t^2}=0. \quad (1.3)$$

The prefactor R/r^2 encodes the cosmological coupling: at $r=R$ the operator reduces to the standard ∂_r^2 on the conformal coordinate, while at atomic scale $r\ll R$ it amplifies the radial derivatives by $R/r^2\sim 10^{22+22}=10^{44}$, which is exactly the factor needed to make $E/\hbar$ and $k^2(r)$ commensurable in the combined equation below.

1.3 Step 3 — Coupling the wave equation to the energy condition

Combining (1.2) and (1.3) and adding the local Coulomb + centrifugal source on the LHS:

$$\boxed{\frac{2m_e}{\hbar^2}\left[E+\frac{e^2}{4\pi\varepsilon_0 r}-\frac{l(l+1)\hbar^2}{2m_e r^2}\right]f(r,t)+\frac{R}{r^2}\frac{\partial^2(r\,f(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}.} \quad (1)$$

This is the **onduscular (TauTimp) electron equation**, in its raw form before separation of variables. With the cosmological radius $R=4.2355\times10^{22}$ m and the boundary condition $g(R)=0$ (where $f(r,t)=g(r)\,u(t)$), it determines the bound spectrum.

1.4 Step 4 — Separation of variables

Let $f(r,t)=g(r)\,u(t)$ with $u(t)=\cos(\omega t)$, so that $\partial_t^2 u/u=-\omega^2$. The energy condition $E=\hbar\omega$ relates the temporal frequency to the electron energy. Substituting into (1):

$$\frac{2m_e}{\hbar^2}\left[E+\frac{e^2}{4\pi\varepsilon_0 r}-\frac{l(l+1)\hbar^2}{2m_e r^2}\right]g(r)+\frac{R}{r^2}\frac{d^2(r\,g(r))}{dr^2}=-\frac{\omega^2}{V^2}\,g(r).$$

The RHS contributes a term $-\omega^2/V^2 \cdot g$ that combines with the energy term on the LHS. For non-relativistic bound states, $V^2 = 2E/m_e$, so $\omega^2/V^2 = E^2 m_e/(2\hbar^2 E) = E m_e/(2\hbar^2)$, which is already absorbed into the $2m_e E/\hbar^2$ term. The radial equation becomes $\frac{R}{r^2}\frac{d^2(r\,g(r))}{dr^2} + g(r)\left[\frac{2m_e E}{\hbar^2} + \frac{2}{a_0 r} - \frac{l(l+1)}{r^2}\right] = 0, \qquad (2)$

where we used $2m_e e^2/(\hbar^2 \cdot 4\pi\varepsilon_0) = 2/a_0$.

This is the **original radial ODE TauTimp** used for the Taylor comparison.

The complete onduscular equation, in the form that reproduces the entire atomic structure of hydrogen, is

$$\boxed{\begin{aligned}&\frac{2m_e}{\hbar^2}\left[E + \frac{e^2}{4\pi\varepsilon_0\, r} - \frac{l(l+1)\hbar^2}{2m_e r^2} + \xi(r)\,\langle \vec{L}\cdot\vec{S}\rangle + V_{HF}(r)\,\langle \vec{S}_e\cdot\vec{I}_p\rangle\right] f(r,t)\\ &\qquad + \frac{R}{r^2}\frac{\partial^2(r\,f(r,t))}{\partial r^2} \;=\; \frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}.\end{aligned}} \qquad (3)$$

with - $\xi(r) = \frac{1}{2m_e^2 c^2 r}\frac{dV}{dr} = \frac{\hbar\alpha}{2m_e^2 c}\frac{1}{r^3}$ for Coulomb $V(r) = -\hbar c\alpha/r$, - $V_{HF}(r) = \frac{2\mu_0}{3}\, g_p\, \mu_B\, \mu_N\, \delta^3(\vec{r}) \cdot \frac{4}{\hbar^2}$ — the Fermi electron–proton contact term.

Equation (3) is the centerpiece of this document.

2. The two governing equations (for comparison with Schrödinger) 2.1 Schrödinger radial equation (hydrogen, reduced form $u = r\psi$)

$$\mathcal{L}_S\, u(r) \;\equiv\; \frac{d^2u}{dr^2} + u(r)\left[\frac{2m_e E}{\hbar^2} + \frac{2}{a_0 r} - \frac{l(l+1)}{r^2}\right] \;=\; 0. \qquad (4)$$

2.2 ODE TauTimp (original operator, scalar, no spin)

From (2):

$$\frac{R}{r^2}\frac{d^2(r\,g(r))}{dr^2} + g(r)\left[\frac{2m_e E}{\hbar^2} + \frac{2}{a_0 r} - \frac{l(l+1)}{r^2}\right] = 0. \qquad (5)$$

With $u(r) = r\,g(r)$, the kinetic term becomes $(R/r^2)\,u''(r)$. Multiplying by r^2/R to clear singularities:

$$u''(r) + \frac{u(r)}{R}\cdot r\left[\frac{2m_e E}{\hbar^2} + \frac{2}{a_0 r} - \frac{l(l+1)}{r^2}\right] = 0.$$

This is the form used in the Taylor comparison below.

3. Taylor comparison at $r = 0$

Multiply (4) by r^2 and (5) (in the $u = rg$ form) by r^2/R to clear $1/r$ and $1/r^2$. Substitute $u(r) = \sum_{n\geq 0} u_n\, r^n$ with the regularity boundary condition $u_0 = 0$ (so that $g(r) = u(r)/r$ is finite at the origin).

3.1 Coefficient table (at $l = 0$, regular solution $u_0 = 0$)

N	$[r^N]$ of Schrödinger $\times\, r^2$	$[r^N]$ of TauTimp $\times\, r^2/R$
0	0 (since $u_0 = 0$)	0
1	$2u_0/a_0 = 0$	$2u_0/(a_0 R)\cdot R = 0$
2	$2u_2 + 2u_1/a_0 + \frac{2m_e E}{\hbar^2}u_0$	$\frac{1}{R}[2u_2 + 2u_1/a_0]$
3	$6u_3 + 2u_2/a_0 + \frac{2m_e E}{\hbar^2}u_1$	$\frac{1}{R}[6u_3 + 2u_2/a_0]$
⋮	⋮	⋮
6	$30u_6 + 2u_5/a_0 + \frac{2m_e E}{\hbar^2}u_4$	$\frac{1}{R}\left[30u_6 + 2u_5/a_0 + \frac{2m_e E}{\hbar^2}u_0\right]$

3.2 Conclusion of the Taylor comparison

- r^0 **coefficient**: equal (both zero) — guaranteed by the regularity condition $u_0 = 0$. ✓
- r^1, r^2: equal trivially because $u_0 = 0$.

- **Divergence at r^3**: Schrödinger has the energy term $\frac{2m_eE}{\hbar^2}u_1$; TauTimp does not (it appears only at r^6 and is additionally suppressed by $1/R$).

Net effect: the original TauTimp ODE misses the energy term $\frac{2m_eE}{\hbar^2}$ at atomic scale; the centrifugal term carries $1/R$ instead of $1/r^2$.

This is the equation for any micro particle that we take for granted. We obtain equation (1) also for v = c, thus m = 0, we have equation (5).
The effective wave function of the onduscular equation has the supposition as null at R (or 2R) is normed, has the dimension of [$L^{-3/2}$], and predicts the probability of the existence of the micro particle in volume dV. Also, a condition of the derivative = R in origin verifies the condition that the speed of interaction has the velocity V, whatever it will be. The value of R = $4.235518529 \cdot 10^{22}$ m, thus 4.477042239 millions of light-years, is calculated in wes.pdf
The change of variable from Schrodinger to QFT Document
Thus, we have:

$$R = \frac{1+\frac{L^2}{r_b\hbar^2}}{8\pi\varepsilon} = \frac{1+\frac{l(l+1)}{r_b}}{8\pi\varepsilon} = \frac{r_b + l(l+1)}{r_b 8\pi\varepsilon} \cong \frac{l(l+1)}{r_b 8\pi\varepsilon}$$

Where rb is the radius Bohr, and ε is the vacuum permittivity, and the value of "l" is the angular momentum quantum number, should be 21.84 (the effective value l = 21.8378 given by the Dirichlet condition g(R) = 0, root n = 11). In the periodic table with 118 elements that we know, the principal quantum number "n" varies from 1 to 5, but theoretically, n should vary from n=1… ∞ (infinity), and the link to "l" is that l varies from 0… n-1. Is a little peculiar that the value of "l" should be close to 21, thus the principal quantum number n = 22; the integer value l = 21 fits the dowsing measurement, and the refined value l = 21.84 gives a proper value to R equal to R = $4.235518529 \cdot 10^{22}$ m, thus 4.477042239 million light-years.

I do not know what the significance of l=21 is that fits the calculus, and a more accurate value, l = 21.84, follows from the Dirichlet condition g(R) = 0 applied to the onduscular equation identified with the Schrödinger equation (see R_l2184_Validare_EN.md), but it is a good start that fills the gap in the theory of a limited Local Universe & the initial conditions for the onduscular theory.

Numerical check at low values (order of Bohr radius) "The Wave Equation Scale",
1. Basic equations (Maple notations: h = ħ)
The Onuscular Equation (eq6 from Maple):

$$\text{eq6} := \frac{2m}{\hbar^2 R}\left(E + \frac{ec^2}{\rho} - \frac{L^2}{2m\rho^2} - \frac{\hbar^2 k}{2m}\right)\rho^2 f(\rho) + 2\frac{d}{d\rho}f(\rho) + \rho\frac{d^2}{d\rho^2}f(\rho) = 0$$

$$\frac{2m}{\hbar^2}\left(E + \frac{ec^2}{\rho} - \frac{L^2}{2m\rho^2} - \frac{\hbar^2 k}{2m}\right)\frac{\rho^2 f(\rho)}{\boldsymbol{R}} + \frac{d^2(\rho f(\rho))}{d\rho^2} = 0$$

Schrödinger equation (eq1 in Maple, section two):

$$\frac{2m}{\hbar^2}\left(E - \frac{ec^2}{\rho}\right)\rho f(\rho) + \frac{d^2(\rho f(\rho))}{d\rho^2} = 0$$

2. Variable change and scaling (rb = Bohr radius)
Substitutions shall apply:

$$\rho = r_b \cdot r, \quad f(\rho) = g(r), \quad E = \frac{m\,ec^4}{32\pi^2\varepsilon^2\hbar^2}E_0, \quad r_b = \frac{4\pi\varepsilon\hbar^2}{m\,ec^2}$$

The scaled Schrödinger equation becomes (eq2, final form):

$$\left(E_0 - \frac{8\pi\varepsilon}{r}\right)r\,g(r) + \frac{d^2(r\,g(r))}{dr^2} = 0$$

Scaled onnduscular equation (onduscular eq2):

$$\left(\frac{E_0 r^2}{R} - \frac{16\hbar^4 k\pi^2\varepsilon^2 r^2}{m^2 ec^4 R} + \frac{8\pi\varepsilon\, r}{R} - \frac{L^2}{\hbar^2 R}\right)rb\cdot g(r) + 2\frac{dg}{dr} + r\frac{d^2 g}{dr^2} = 0$$

3. Changing variables with the limit factor (border condition)
The zero transformation to r = 2R is applied:

$$f(\rho) = g(r) \cdot \frac{R - r \cdot r_b}{2R}$$

After substitution and simplification (eq2 → eq3 in Maple), the equation becomes:

$$r\frac{d^2g}{dr^2} + 2\frac{dg}{dr} + \mathcal{T}(r) \cdot g(r) = 0$$

where the term free expands in Taylor series around :$\mathcal{T}(r)r = 0$

$$\mathcal{T}(r) = \underbrace{\frac{r_b(8\pi R\varepsilon - 2)}{2R}}_{\text{ord. 0}} + \underbrace{\frac{r_b(E_0 R\, r_b - 8\pi\varepsilon\, r_b) + \frac{r_b^2(8\pi R\varepsilon - 1)}{R}}{2R} \cdot r}_{\text{ord. 1}} + \underbrace{\frac{-r_b^3 E_0 + \frac{r_b^3(2E_0R^2 - 1)}{R^2}}{2R} \cdot r^2}_{\text{ord. 2}}$$

4. The condition of overlap with Schrödinger

Comparing the coefficients with the scaled onuscular equation:

$$\frac{E_0 r^2}{R} - \frac{16\hbar^4 k\pi^2\varepsilon^2 r^2}{m^2 e c^4 R} + \frac{8\pi\varepsilon r}{R} - \frac{L^2}{\hbar^2 R}$$

3 equations are obtained by identifying the coefficients at each order in: r

Order	Matching equation
r^0 (constant)	$\frac{r_b(8\pi R\varepsilon - 2)}{R} = -\frac{L^2}{\hbar^2 R}$
r^1 (linear)	$\frac{r_b(2E_0 R\, r_b - 8\pi\varepsilon r_b) + \frac{r_b^2(8\pi R\varepsilon - 1)}{R}}{R} = -\frac{8\pi\varepsilon}{R}$
r^2 (quadratic)	$\frac{1}{2} \cdot \frac{-r_b^3 E_0 + \frac{r_b^3(2E_0R^2 - 1)}{R^2}}{R} = \frac{E_0}{R} - \frac{16\hbar^4 k\pi^2\varepsilon^2}{m^2 e c^4 R}$

5. Analytical solution wes.mw (31)

Maple solves the system of 3 equations in and gets: R, k $\boxed{R = -\frac{2r_b\hbar^2 - L^2}{8\pi\varepsilon\hbar^2 r_b}}$

or equivalent (eq. 32 of Maple): $R = \frac{-r_b h^2 + L^2}{8\pi\varepsilon h^2 r_b}$

where h = ħ (Maple notation).

The simplified form (Maple equation 37), for the orbital quantum number:l $\boxed{R = \frac{l(l+1)}{8\pi\varepsilon\, r_b}}$

This results from the identity, as follows:$L^2 = l(l+1)\hbar^2$

$R = \frac{r_b\hbar^2 + l(l+1)\hbar^2}{8\pi\varepsilon\hbar^2 r_b} = \frac{l(l+1) + 1\cdot(r_b/r_b)}{8\pi\varepsilon\, r_b} \approx \frac{l(l+1)}{8\pi\varepsilon\, r_b} \quad (l \gg 1)$

A more rigorous calculation of the value of the constant distance R in the supposition of identity of spectral lines of Hydrogen, in the Schrodinger model, and the onduscular equation, so that the error corresponds to a precision $\varepsilon \approx 10^{-64}$ — 62 orders smaller than the finest spectroscopic measurement.

$$\boxed{R(n_1, n_2, \varepsilon) = \frac{\pi\, a_0\, n_1\, n_2}{\sqrt{\varepsilon}}}$$

$$R = \frac{\pi\, a_0\, n_1 n_2}{\sqrt{\varepsilon}}$$

$$R_{\text{opt}}(\varepsilon_{\text{rms}}) = \frac{\pi\, a_0 \sqrt{\langle n_1^2 n_2^2\rangle}}{\varepsilon_{\text{rms}}}$$

With : $\sqrt{\langle n_1^2 n_2^2\rangle} = 9.959$, giving:

$$R_{\text{opt}}(\varepsilon_{\text{rms}}) = \frac{\pi \cdot a_0 \cdot 9.959}{\varepsilon_{\text{rms}}} = \frac{1.656 \times 10^{-9}\ \text{m}}{\varepsilon_{\text{rms}}}$$

So, we have R ≈ 4.2355·10^{22} m link R.pdf

21 May 2026 (updated 7 November 2026: l = 21.84, R = 4.2355·10^{22} m)